Marker Beacon – General Data

1. General

The Marker Beacon (MKR) uses visual and audio indicators to alert the crew for when the aircraft passes over a 75 MHz transmitter. The Marker Beacon receiver is contained in the audio control panel (KMA 29) (refer to 23-53-00, Audio Control Panel – General Data).

2. Description

A. Marker Beacon

Marker Beacon receive signals are sent to the Modular Avionics Unit (MAU) and then displayed on the Primary Flight Display (PFD). The Outer Marker has an associated 400-Hertz ‘dash' tone. The tone will be keyed at a rate of two tones per second when the aircraft is in the range of the Outer Marker Beacon. The Middle Marker is coupled with a 1300-Hertz tone. It is keyed alternately with short ‘dot' and long ‘dash' bursts at 95 combinations per minute. The Inner Marker has a 3000-Hertz ‘dot' tone. The tone will be keyed at a rate of six times per second.

The audio from the Marker Beacon receiver can be heard by selecting the “MKR” push-button switch. To adjust the volume level, there is a service adjustment located on the unit. The Marker Beacon CI 118 antenna is installed on the bottom of the fuselage, right of the centre line, between station 73.00 and 86.00.

3. Adjustment/Test

A. Function Test

Required Equipment:

- IFR 4000 or equivalent
- Milliohm or Micro-Ohmmeter (AEMC Micro-Ohmmeter Model 6250 or similar – required accuracy: 0.0005 Ω)

1) General

These procedures are for the function test of the Marker Beacon.

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that PILOT AUDIO (3E) and COPILOT AUDIO (4E) circuit breakers are closed.

d) Press and hold the MKR MUTE/TEST button on the pilot audio control panel.

e) Verify the inner marker indication (white boxed I) is displayed on the PFD.

f) Set the test set to generate each of the following:

- 400 Hz tone for Outer Marker (blue boxed ‘O’)
- 1300 Hz tone for Middle Marker (amber boxed ‘M’)
- 3000 Hz tone for Inner Marker (white boxed ‘I’)

Note For each generated tone, make sure the respective marker indication is displayed on both PFD’s.

g) Remove test set.

h) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Audio Control Panel Marker Beacon Audio Adjustment

1) General

These procedures are for Marker Beacon audio level adjustment.

2) Adjustment

a) Ensure that PILOT AUDIO (3E) circuit breaker is closed.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Wait several minutes for the aircraft electrical systems to initialize.

d) Set up the test equipment to generate a 400, 1300 or 3000 Hertz tone.

e) Connect a headset to the pilot position.

f) Select MKR push button switch on the pilot audio control panel and ensure the 400, 1300, or 3000 Hertz tone is heard.

g) Evaluate the audio level.

h) Remove the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 4.A.2).

i) Adjust the pilot's audio panel MRK potentiometer adjustment on the top of the unit.

j) Install the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 4.A.3).

k) Test the pilot's audio control panel as per 23-53-00, Audio Control Panel – General Data, Para 3.A.

l) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

C. Electrical Bonding Test – Marker Beacon Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to right-hand Access and Inspection Provision 15 panel. (Refer to 12-00-00, Servicing – General Data).

3) Turn DZUS fastener 1/4 turn and swing open access panel.

4) Locate Marker Beacon Antenna aft of station 73.0.

5) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the Marker Beacon Antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be 3.0 milliohms unless stated otherwise.

6) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

7) If the electrical conductivity is not consistent over the electrical bond contact area check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

8) Close right-hand Access and Inspection Provision 15 and tighten the DZUS fasteners.

4. Removal and Installation

A. Removal and Installation of the Marker Beacon

The Marker Beacon receiver is contained in the Audio Control Panel (KMA 29). Refer to 23-53-00, Audio Control Panel – General Data Para 4. for Removal/Installation of the Audio Control Panel (KMA 29).

B. Removal and Installation of the Marker Beacon Antenna (CI 118)

1) General

These procedures are for the removal and installation of the Marker Beacon antenna (CI 118).

2) Removal

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Pull the PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Carefully cut the sealant from around the base of the antenna.

d) Remove the screws (6) from the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

Note Stow connectors of the removed components.

3) Installation

a) Ensure the aircraft power is OFF (refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Solvent clean the exposed area of the airframe around the antenna (refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Bring the antenna assembly close to the fuselage, connect the coaxial connector.

e) Install the screws (6) securing the antenna to the fuselage.

f) Check the antenna electrical bonding as per Para 3.C.

g) Remove safety clips and close PILOT AUDIO (3E) and/or the COPILOT AUDIO (4E) circuit breaker on the avionics circuit breaker panel.

h) Test the Marker Beacon as per Para 3.A.

i) Touch up the exposed area of airframe around the antenna (if required) with F19 primer by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

j) Use PR-1422 sealant to both fillet seal around the antenna base and dome seal the securing screws. The aircraft will be ready to fly approximately 24 hours after sealing.

5. Troubleshooting

The Marker Beacon receiver is contained in the Audio Control Panel (KMA 29). Refer to 23-53-00, Audio Control Panel – General Data, Para 5. for Troubleshooting of the Audio Control Panel (KMA 29).